Software Design Specifications

for

Peer-to-Peer University Tutoring Platform with Personalized Recommendation System

Prepared by

Yukta Menon	se23ucse192
Sujan Vadla	se23ucse178
Hridya Reddy	se23uari066
Pranishk Reddy	se23uari099
Bharat Reddy	se23uari020

Instructor:	**Avinash Chauhan**
Course:	**Software engineering**
Group number:	***43***
Teaching Assistant:	**Sai Srithaja**
Date:	07/04/2026

Document Information

Title: Peer-to-peer university tutoring platform with recommender system	
Project Manager:yukta menon	**Document Version No:**
	Document Version Date:
Prepared By:Pranishk reddy	**Preparation Date:07/04/2026**

Version History

Ver. No.	Ver. Date	Revised By	Description	Filename

Table of Contents

1 INTRODUCTION 4

1.1 PURPOSE 4
1.2 SCOPE 4
1.3 DEFINITIONS, ACRONYMS, AND ABBREVIATIONS 4
1.4 REFERENCES 4

2 USE CASE VIEW 4

2.1 USE CASE 4

3 DESIGN OVERVIEW 4

3.1 DESIGN GOALS AND CONSTRAINTS 5
3.2 DESIGN ASSUMPTIONS 5
3.3 SIGNIFICANT DESIGN PACKAGES 5
3.4 DEPENDENT EXTERNAL INTERFACES 5
3.5 IMPLEMENTED APPLICATION EXTERNAL INTERFACES 5

4 LOGICAL VIEW 5

4.1 DESIGN MODEL 6
4.2 USE CASE REALIZATION 6

5 DATA VIEW 6

5.1 DOMAIN MODEL 6
5.2 D ATA MODEL (PERSISTENT DATA VIEW) 6
5.2.1 Data Dictionary 6

6 EXCEPTION HANDLING 6

7 CONFIGURABLE PARAMETERS 6

8 QUALITY OF SERVICE 7

8.1 AVAILABILITY 7
8.2 SECURITY AND AUTHORIZATION 7
8.3 LOAD AND PERFORMANCE IMPLICATIONS 7
8.4 MONITORING AND CONTROL 7

1 Introduction

[The introduction of the Software Design Specifications should provide an overview of the entire Software Design. It should include the purpose, scope, definitions, acronyms, abbreviations, references, and overview of the Software Design Specifications.]

1.1 Purpose

The purpose of this document is to describe the software design of the Peer-to-Peer University Tutoring Platform with a Personalized Recommendation System. It serves as a reference for developers, testers, and stakeholders to understand system architecture, components, and interactions.

1.2 Scope

This system enables university students to act as both tutors and learners. Users can search for tutors, book sessions, and receive personalized tutor recommendations using machine learning techniques based on preferences and past interactions.

1.3 Definitions, Acronyms, and Abbreviations

- P2P – Peer-to-Peer
- ML – Machine Learning
- API – Application Programming Interface
- UI – User Interface

1.4 References

- IEEE Software Engineering Standards
- Research papers on Recommender Systems
- Online documentation for APIs (Stripe, WebRTC)

2 Use Case View

[Based on the requirements document, this section lists use cases or scenarios from the use-case model if they represent some significant, central functionality of the final system, or if they have a large design coverage - they exercise many design elements, or if they stress or illustrate a specific, delicate point of the design. Provide a use case diagram for use cases that pertains to the software design]

2.1 Use Case

Actors:

- Student (Learner)
- Tutor
- Admin

Main Use Cases:

- Register/Login
- Search Tutor
- View Tutor Profile
- Book Session
- Make Payment
- Attend Session
- Rate & Review Tutor
- Get Personalized Recommendations

3 Design Overview

[This section provides an overview of the entire software design. It references and complies with the high-level design interface contracts, requirements and high-level module decomposition approach.]

3.1 Design Goals and Constraints

- Fast and accurate tutor matching
- User-friendly interface
- Scalable architecture
- Secure data handling

3.2 Design Assumptions

- Users have internet access
- Devices support video calls
- Payment systems are functional

3.3 Significant Design Packages

- Authentication Module
- Tutor Matching Module
- Recommendation Engine (ML Module)
- Booking & Scheduling Module
- Payment Module
- Notification Module

3.4 Dependent External Interfaces

- Payment Gateway (Stripe/Razorpay)

- Video Communication API (WebRTC/Zoom SDK)

External Application and Interface Name	Module Using the Interface	Functionality/ Description
Payment Gateway (Razorpay/Stripe)	Payment Module	Used to process tutor session payments securely

The table below lists the public interfaces this design requires from other modules or applications.

3.5 Implemented Application External Interfaces (and SOA web services)

- User Authentication API
- Booking API
- Recommendation API

The table below lists the implementation of public interfaces this design makes available for other applications.

Interface Name	Module Implementing the Interface	Functionality/ Description
Recommendation API	Recommendation Engine	Returns personalized tutor suggestions using ML

4 Logical View

[This section describes the detailed design. The description should consist of a series of layers. The top most layers should show how the application modules interact to complete the key use cases from Section 2. The more detailed layers should expand each module's separate interaction responsibility and show how the separate classes within the

module collaborate to implement the required behaviour. Depending on project needs, this decomposition may continue until each method in the layer can be described in a page or less of pseudo-code or text.]

4.1 Design Model

Main Classes:

- User (Base Class)
- Student (inherits User)
- Tutor (inherits User)
- Session
- Review
- RecommendationEngine

Relationships:

- A Student books Sessions
- A Tutor conducts Sessions
- Sessions generate Reviews
- RecommendationEngine suggests Tutors

4.2 Use Case Realization

1. User logs in
2. Student searches subject
3. System retrieves tutor list
4. Recommendation engine ranks tutors
5. Student selects tutor
6. Booking request sent
7. Tutor accepts/rejects
8. Payment processed
9. Session scheduled

5 Data View

[This section describes the persistent data storage perspective of the system. This section is may not apply if there is little or no persistent data, or the translation between the Design Model and the Domain Model is trivial]

5.1 Domain Model

Entities:

- User
- Tutor
- Student
- Session
- Review

5.2 Data Model (persistent data view)

Table	Fields
User	UserID, Name, Email, Role
Tutor	Subjects, Rating, Experience
Student	Preferences
Session	SessionID, Time, Status
Review	Rating, Feedback

5.2.1 Data Dictionary

- UserID → Unique identifier
- Rating → Numeric score (1–5)
- SessionStatus → Booked/Completed/Cancelled

6 Exception Handling

- No tutors available → Show message
- Payment failure → Retry option
- Session conflict → Reschedule
- Network error → Notify user

7 Configurable Parameters

This table describes the simple configurable parameters (name / value pairs).

Parameter	Definition	Dynamic
Max Session Duration	Limits session time	Yes
Recommendation Weight	Controls ranking logic	Yes
Rating Threshold	Minimum tutor rating	Yes

8 Quality of Service

[This section describes aspects of the design related to application availability, security, performance, and monitoring and control in production.]

8.1 Availability

- System available 24/7
- Cloud-based deployment

8.2 Security and Authorization

- Secure login (JWT/Auth tokens)
- Encrypted data transmission
- Role-based access control

8.3 Load and Performance Implications

- Handles multiple concurrent users
- Tutor search response < 2 seconds